Supplementary file for "Robust Fault Diagnosis for Cyber-Physical Systems Under Sensor-Reading Modification Attacks via Dynamic Observation"

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This file provides a complete proof for Theorem 3 in manuscript "Robust Fault Diagnosis for Cyber-Physical Systems Under Sensor-Reading Modification Attacks via Dynamic Observation".

Theorem 3: Given plant G with Σ_f, a set of vulnerable events $\Sigma_v \subseteq \Sigma_o$, and D, let G_t^a be the test automaton constructed from D under SM attacks using *ConstructGta*. G is robustly diagnosable w.r.t. Σ_f, Σ_v, and D iff G_t^a contains no NSCC.

Proof: We first clarify certain properties that are necessary for the proof. Let $X_N = X \times \{N\}$ and $X_Y \times \{Y\}$. For a path $v \in L(G_t^a)$, $P_G(v)$ denotes the actual string generated by G. Let $m(v) \in (\Theta \cup \Sigma_o)^*$ denote the modified run received by D w.r.t. v. Specifically, $\forall s \in (\Theta \cup \Sigma \cup (\Sigma_o \times \Sigma_o))^*$, and $e \in (\Theta \cup \Sigma \cup (\Sigma_o \times \Sigma_o))$,

$$m(se) = \begin{cases} m(s)\theta, & \text{if } e = \theta \in \Theta \\ m(s)\sigma', & \text{if } e = (\sigma, \sigma') \in (\Sigma_o \times \Sigma_o) \\ m(s)\sigma, & \text{if } e = \sigma \in \theta' \\ m(s), & \text{if } e = \sigma \notin \theta' \end{cases},$$

where θ' is the last sensing decision in s. According to construct of G_t^a, we have $\eta_t^a(z_{0,t}^a, v) = (\tilde{\eta}_D(\tilde{z}_{0,D}, m(v)), \delta_l(x_{0,l}, P_G(v)))$. Moreover, there exists a run $w \in L(D)$ such that $P_G(v) \in P_\Theta^{-1}(w)$ and $m(v) \in M_\Theta(w)$. In addition, we have $\tilde{\eta}_D(\tilde{z}_{0,D}, m(v)) = risk \Leftrightarrow m(v) \notin L(D)$. Based on the transition function of D, $\forall w \in L(D)$, if $\eta_D(z_{0,D}, w) = z$ or (z, θ), then $z = \{x_l \in X_l | \exists s \in P_\Theta^{-1}(s) \cap L(G),\ \delta_l(x_{0,l}, s) = x_l\}$. Since $s \in L_N \Leftrightarrow \delta_l(x_{0,l}, s) \in X_N$, which indicates that

$$z \cap X_N \neq \varnothing \Leftrightarrow P_\Theta^{-1}(m(w)) \cap L_N \neq \varnothing. \quad (1)$$

We now prove the two directions.

($\Rightarrow$) By contradiction, assume that G_t^a contains an NSCC, which involves a decision state (z, θ, x_l), where

$$x_l \in X_Y \text{ and } z \cap X_N \neq \varnothing. \quad (2)$$

Let $v \in L(G_t^a)$ and $\eta_t^a(z_{0,t}^a, v) = (z, \theta, x_l)$. There exists $v' \in (\Theta \cup \Sigma \cup (\Sigma_o \times \Sigma_o))^* \backslash \{\varepsilon\}$, such that $\eta_t^a((z, \theta, x_l), v') = (z, \theta, x_l)$. Based on transition rule of G_t^a, $|P_G(v')| > 0$. Let $v_n = v(v')^n$, where $n \in \{1, 2, \ldots\}$. Since $x_l \in X_Y$, a fault has occurred. $P_G(v_n)$ can be written as $P_G(v_n) = st_n$, where $s \in \Psi(\Sigma_f)$ and t_n can be made arbitrarily large by increasing n. According to construction of D, there exists $w_n \in L(D)$ such that $st_n \in P_\Theta^{-1}(w_n)$ and $m(v_n) \in M_\Theta(w_n)$. As $\eta_t^a(z_{0,t}^a, v_n) \notin \{risk\} \times X_l$, we have $m(v_n) \in L(D)$. Based on (1) and (2), we have $P_\Theta^{-1}(m(v_n)) \cap L_N \neq \varnothing \Rightarrow [P_\Theta^{-1}(m(v_n)) \cap L(G)] \nsubseteq L_F$. Based on Definition 6, G is not robustly diagnosable, which causes contradiction.

($\Leftarrow$) By contradiction, suppose that G is not robustly diagnosable. By Definition 6, $\exists s \in \Psi(\Sigma_f)$, $t \in L(G)/s$ with $|t| > k$ for any arbitrarily large k, and $\exists w \in L(D)$, $st \in P_\Theta^{-1}(w)$, and $u \in M_\Theta(w)$, such that $u \in L(D)$ and $[P_\Theta^{-1}(u) \cap L(G)] \nsubseteq L_F$. According to construction of G_t^a, $\exists v_f v_t \in L(G_t^a)$, such that $s = P_G(v_f)$, $t = P_G(v_t)$, and $u = m(v_f v_t)$. Since $st \in L_F$ and $u \in L(D)$, $\delta_l(x_{0,l}, st) \in X_Y$ and $\eta_t^a(z_{0,t}^a, v_f v_t) \notin \{risk\} \times X_l$. Let $\eta_t^a(z_{0,t}^a, v_f v_t) = (z, x_l)$ or (z, θ, x_l). By (1), we have $z \cap X_N \neq \varnothing$. Since G_t^a is a finite state automaton, any path in G_t^a contains at most $|Z_t^a|$ different states. As $|t|$ can be arbitrarily large, we have $|t| > |Z_t^a| \Rightarrow |v_t| > |Z_t^a|$. $v_f v_t$ ultimately leads to a nontrivial SCC in G_t^a. It implies that $\exists v_{t1}, v_{t2}, v_{t3} \in (\Theta \cup \Sigma \cup (\Sigma_o \times \Sigma_o))^*$ with $v_{t2} \neq \varepsilon$, and $(z, \theta, x_l) \in Z_t^a$, such that $v_t = v_{t1}v_{t2}v_{t3}$, $\eta_t^a(z_{0,t}^a, v_f v_{t1}) = (z', \theta', x_l')$, and $\eta_t^a((z', \theta', x_l'), v_{t2}) = (z', \theta', x_l')$. Accordingly, $\exists t_1, t_2, t_3 \in \Sigma^*$, such that $t = t_1 t_2 t_3$, $t_1 = P_G(v_{t1})$, $t_2 = P_G(v_{t2})$, and $t_3 = P_G(v_{t3})$. We have $x_l' = \delta_l(x_{0,l}, st_1t_2) \in X_Y$. We then show that $z' \cap X_N \neq \varnothing$. If $z' \cap X_N = \varnothing$, then $z' \subseteq X_Y$ and $\eta_t^a((z', \theta', x_l'), v_{t3}) = (z, x_l)$ or (z, θ, x_l), where $z \subseteq X_Y$. Clearly, it contradicts the fact that $z \cap X_N \neq \varnothing$. Thus, $z' \cap X_N \neq \varnothing$ and $x_l' \in X_Y \Rightarrow$ the nontrivial SCC that involves (z', θ', x_l') is an NSCC, which contradicts that G_t^a contains no NSCC. ♦